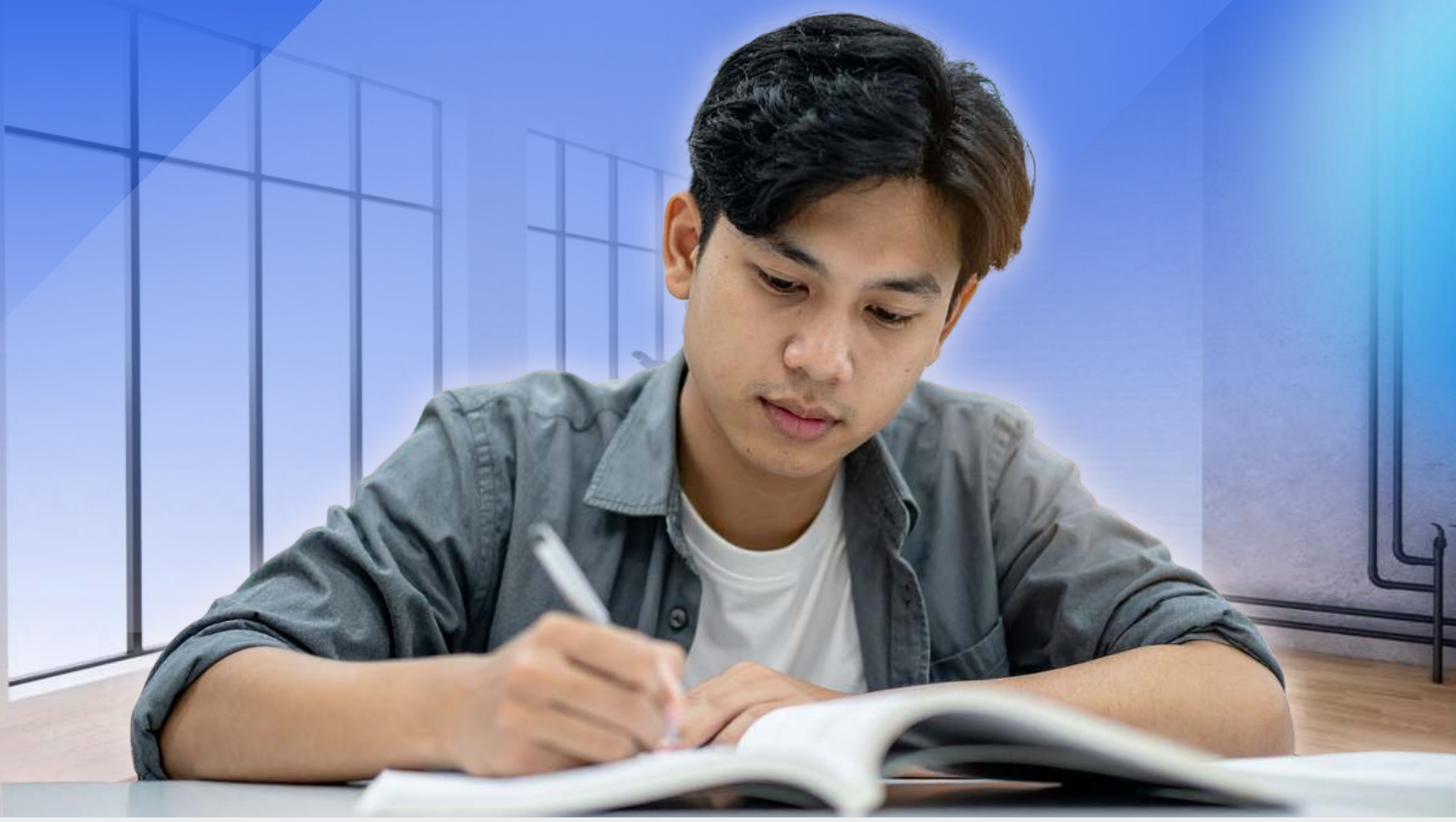
Lenterago

# TRY OUT #11

## SAT-Math

## 61011

44 QUESTIONS



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# MODUL 1 & 2

22 QUESTIONS

DIRECTIONS

The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

Unless otherwise indicated:

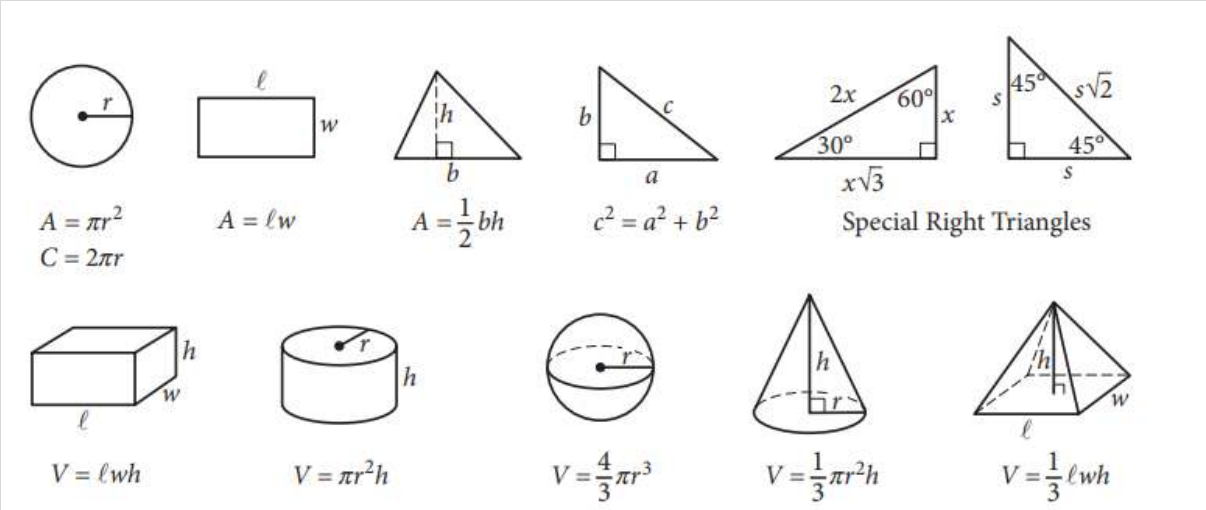
A. All variables and expressions represent real numbers.

B. Figures provided are drawn to scale.

C. All figures lie in a plane.

D. The domain of a given function *f* is the set of all real numbers *x* for which *f(x)* is a real number

REFERENCE




The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2p.

The sum of the measures in degrees of the angles of a triangle is 180.



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# Module 1 & 2

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer next to or under the question in the test book as described below.

1. Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle, or for any questions with more than one circled answer.
2. If you find more than one correct answer, write and circle only one answer.
3. Your answer can be up to 5 characters for a positive answer and up to 6 characters (including the negative sign) for a negative answer, but no more.
4. If your answer is a fraction that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
5. If your answer is a decimal that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
6. If your answer is a mixed number (such as 3 1/2), write it as an improper fraction (7/2) or its decimal equivalent (3.5).
7. Don't include symbols such as a percent sign, comma, or dollar sign in your circled answer.



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| NO | MATERI | SUB MATERI | KUNCI |
|---|---|---|---|
| 1 | Algebra | | |
| | If $m = 4$, how much greater 10m than 6m ?<br>a. 4<br>b. 6<br>c. 16<br>d. 20 | | |
| 2 | Algebra | Word problem of SLE | |
| | The sum of the two different number a and b is 70 and the difference when the smaller number is subtracted from the larger number is 30. What is the value of $ab$?<br>a. 100<br>b. 210<br>c. 1000<br>d. 2100 | | |
| 3 | Algebra | Line | |
| | Line $l$ intersects the graph of the function $f(x) = 2x^2 - 4x + 1$ at two point where $x = -1$ and $x = 2$, respectively. What is the slope of line $l$?<br>a. $-2$<br>b. $-\frac{2}{3}$<br>c. $\frac{3}{2}$<br>d. 2 | | |
| 4 | Algebra | Word Problem | |

| Section | Price per ticket | Number of Sold |
|---|---|---|
| Front Orchestra | \$60 | 50 |
| Rear Orchestra | \$50 | 60 |
| First Mezzanine | \$40 | $x$ |
| Second Mezzanine | \$35 | $y$ |
| Third Mezzanine | \$30 | 100 |

The table above show information about tickets sold for a recent performance by a theater troupe. The total revenue in ticket sales for this performance is \$15.000.
If 15 more tickets were sold in the second mezzanine than in the first mezzanine, what is the total number of the ticket that were sold for this performance?

a. 300
b. 318
c. 371
d. 380

| 5 | Algebra | Word problem | |
|---|---|---|---|
| | $d = 2,565 - 500t$ | | |



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An airplane flies directly from a city in UK to a city in USA. The equation above estimates the distance $d$, in miles, from the city in USA of the plane after $t$ hours after taking off from the city in the UK.
Which of the following is the best interpretation of the number $2,565$ in this context?

a. The speed, in miles per hour, of the air plane
b. The distance, in miles, the air plane travels in one hour
c. The distance , in miles, the air plane travel between the two cities
d. The time, in minutes, it take the airplane to reach the city in USA

| 6 | Algebra | Inequality word Problem | |
|---|---|---|---|

David mows grass at a constant rate of 1.5 acres per hour. He mowed 2 acres before lunch and plans to spend $t$ hours mowing after lunch. If David wants to mow at least 8 acres of grass today, which of the following inequalities best represents this situation?

a. $1,5t \geq 8$
b. $1.5t - 2 \geq 8$
c. $1.5t + 2 \geq 8$
d. $2t + 1.5 \geq 8$

| 7 | Algebra | Linear Function | |
|---|---|---|---|

| $x$ | $h(x)$ |
|---|---|
| -1 | 1 |
| 2 | 7 |
| 4 | 11 |

The table above shows the selected values for the function h. In the xy-plane, the graph of $y = h(x)$is a line. What is the value of $h(8)$?

a. 15
b. 19
c. 21
d. 22

| 8 | Advanced Math | Quadratic | |
|---|---|---|---|

Which the following equation represent a parabola in the $xy$ −plane with a vertex that lies on the x-axis?

a. $y = (x - 2)^2 + 2$
b. $y = 2(x + 3)^2$
c. $y =- 2x^2 + 3$
d. $y = 3x^2 + 2$

| 9 | Advanced Math | Quadratic Equation | |
|---|---|---|---|

The graph of quadratic function is $y =- x^2 + 4x + 12$ has two $x$ −intercepts at point A and B. The vertex of the parabola is point C. What is the area of triangle ABC?

a. 32
b. 48
c. 64


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| | | | |
|---|---|---|---|
| | d. 75 | | |
| 10 | Advanced Math | Transformation of function | |
| | In the $xy$ −plane, the graph of $y = (x - 6)^2 + 3$ is the image of the graph of $y = (x + 7)^2 + 3$ after a translation.<br>Which the following statement is true?<br>a. Shift to the left 13 unit<br>b. Shift to the right 13 unit<br>c. Shift to the left 1 unit<br>d. Shift to the right 1 unit | | |
| 11 | Advanced Math | Rational Expression | |
| | What is the possible sum of solution to the equation<br>$\frac{6}{x+1} - \frac{3}{x-1} = \frac{1}{4}$ | | |
| 12 | Advanced Math | Exponent | |
| | If $y = x^2$ where $x \neq 0$ and $w = y^6$ which of the following expresses the value of $\frac{w}{y^3}$ in the term of x?<br>a. $x^2$<br>b. $x^4$<br>c. $x^5$<br>d. $x^6$ | | |
| 13 | Advanced Math | Exponential Word Problem | |
| | $V(t) = 1000(1 + k)^m$<br>An analyst wants to use the formula above to estimate the value, in dollars, of a \$1,000 initial investment in a mutual fund after $m$ quarter have passed. If a \$1,102.50 after 2 quarters, what is the number of $k$?<br>a. 0.05<br>b. 0.25<br>c. 0.5<br>d. 0.65 | | |
| 14 | Advanced Math | Exponent | |
| | Which of the following expression is equivalent to $(16x^9y^3)^{\frac{1}{2}}$ , where $x \geq 0$ and $y \geq 0$?<br>a. $4x^3y^{\frac{3}{2}}$<br>b. $4x^{\frac{9}{2}}y^{\frac{3}{2}}$<br>c. $8x^3y^3$<br>d. $8x^{\frac{9}{2}}y^3$ | | |



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| | | | |
|---|---|---|---|
| 15 | Problem Solving and Data Analysis | Rate and Percent | |

After its initial offering, the price of a stock increased by 20% in the first year, decreased by 25% in the second year, then increased by 10% in the third year. What was the net change in the stock price over the entire three three-year period?

a. It increased by 5%
b. It increased by 1%
c. In decreased by 1 %
d. It decreased by 5 %

| | | | |
|---|---|---|---|
| 16 | Problem Solving and Data Analysis | Rate | |

A state environmental study determined that a coastal region of the state lose 24.5 acres of wetlands per month. At this rate, how many months will I take coastal region to lose a total 343 acres

| | | | |
|---|---|---|---|
| 17 | Problem Solving and Data Analysis | Median | |

Monthly Enrollments in Swimming Training Classes

| Community Center | Jan | Feb | Mar | Apr | May | Jun | Jul |
|---|---|---|---|---|---|---|---|
| A | 124 | 111 | 98 | 82 | 77 | 152 | 133 |
| B | 465 | 407 | 391 | 354 | 365 | 511 | 495 |

The table above shows monthly enrollment in swimming training classes at two community for 7 consecutive months. Based on the table, what is the difference of median monthly enrollment between two community?

| | | | |
|---|---|---|---|
| 18 | Problem Solving and Data Analysis | Standard Deviation of Data | |

| Data Set A | Data Set B |
|---|---|
| 25,550 | 22,860 |
| 40,430 | 55,020 |
| 49,150 | 173,730 |
| 62,590 | 300,560 |
| 73,670 | 358,930 |
| 118,780 | 465,170 |
| 126,040 | 600.430 |

Which of the following is true about the standard deviation of the two data set in the table above?

a. The standard deviation of data B is larger than the standard deviation data set A
b. The standard deviation of data A is larger than the standard deviation data set A
c. The standard deviation of data A is equal to the standard deviation data set A
d. There is not enough information available to compare the standard deviation of the two data sets

| | | | |
|---|---|---|---|
| 19 | Problem Solving and Data Analysis | Probability | |

A sample of 600 ninth graders was selected at random and asked how much time they spend on homework each day. Of the ninth graders selected, 220 spend less than 2 hours on homework each day. If the conclusion was drawn that "approximately 1.35 million ninth graders spend less than 2 hours on homework each day," which of the following is closest to the population, in millions, of ninth graders?

a. 0.495


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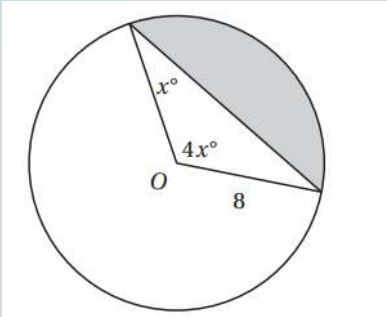
| | | | |
|---|---|---|---|
| | b. 1.35<br>c. 3.68<br>d. 5.84 | | |
| 20 | Geometry | Area of Circle | |
| |  <br>Point O is the center of the circle above. What is the area of the shaded region?<br>a. $\frac{64\pi}{3} - 16\sqrt{3}$<br>b. $\frac{16\pi}{3} - 8\sqrt{3}$<br>c. $\frac{64\pi}{3} - 12\sqrt{3}$<br>d. $\frac{64\pi}{3} - 8\sqrt{3}$ | | |
| 21 | Geometry | Volume | |
| | The volume of sphere A is 20 cubic centimeters. Sphere B has radius that is 3 time of the radius of sphere A. What was the volume , in cubic centimeter of sphere B? | | |
| 22 | Geometry | Equation of Circle | |
| | Which of the following equation describe a circle with radius 10 passes through the origin when graphed in the $xy$ −plane?<br>a. $(x - 10)^2 + (y + 10)^2 = 100$<br>b. $(x - 5)^2 + (y - 5)^2 = 100$<br>c. $(x - 5\sqrt{2})^2 + (y + 5\sqrt{2})^2 = 100$<br>d. $x^2 + y^2 = 100$ | | |



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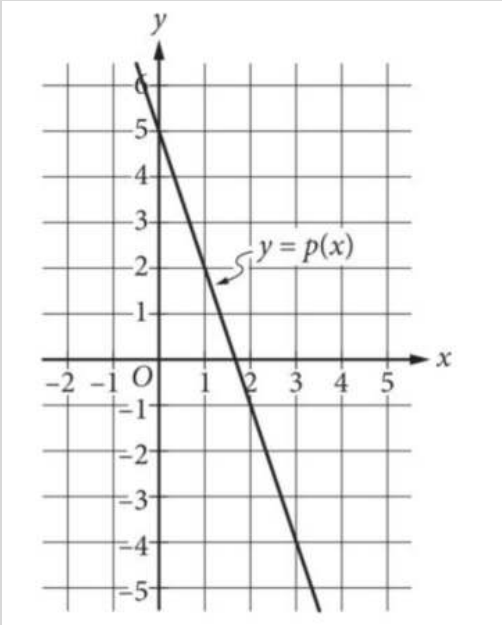
| NO | MATERI | SUB MATERI | KUNCI |
|---|---|---|---|
| 1 | Algebra | Linear Equation | |

If $\frac{6}{x+1} = 3$ what is the value of 2+2x?

a. 0,5
b. 1
c. 2
d. 4

| 2 | Algebra | Linear Function + Transformation | |
|---|---|---|---|




The graph of linear equation $p(x)$ is shown above. The line $p(x)$ is reflected to y axis and become line $r(x)$. Which equation define $r(x)$?

a. $r(x) = -3x + 5$
b. $r(x) = 3x + 5$
c. $r(x) = 3x - 5$
d. $r(x) = \frac{1}{3}x + 5$

| 3 | Algebra | Word Problem | |
|---|---|---|---|

One bag of grass seed can cover 4,000 square feet of new lawn. If each bag cost p dollar, which of the following expression give the cost, in dollars, to cover a new rectangular lawn that measure a feet by b feet?

a. $\frac{4{,}000p}{ab}$
b. $\frac{abp}{4{,}000}$
c. $\frac{4{,}000ab}{p}$
d. $4,000abp$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| | | | |
|---|---|---|---|
| 4 | Algebra | Linear Equation word problem | |

$$D = 60 - \frac{3}{4}P$$
$$S = \frac{1}{4}P$$

In economics, the equilibrium price is defined as the price at which quantity demanded and quantity supplied are equal. If the quantity demanded, D, and quantity supplied, S, in the term of price in dollars, P, are given by equation above, what is the equilibrium price?

a. \$10
b. \$60
c. \$80
d. \$120

| | | | |
|---|---|---|---|
| 5 | Algebra | Linear Line | |

In the xy-plane , line m has a slope of 2 and crosses the x-axis at the point $(\frac{5}{2}, 0)$. Line n is perpendicular to line m and crosses the y axis at $\left(0, -\frac{5}{2}\right)$. At what point do line m and n intersect?

a. $(\frac{5}{2}, -\frac{5}{2})$
b. $4, -2)$
c. $(1, -3)$
d. $(2, -\frac{5}{2})$

| | | | |
|---|---|---|---|
| 6 | Algebra | Word problem | |

A geographer found that the land area of Aruba is 25 square miles and the land area of Bermuda is 9 square miles. Based on these data, the geographer used the equation 25x+9y = T to estimate the total number of residents, T, of the two islands. Which is the best interpretation of x in this context?

a. The total number of residents of Aruba
b. The number of square miles of Aruba
c. The total number of residents of both islands
d. The population density, in residents per square mile, of Aruba.

| | | | |
|---|---|---|---|
| 7 | Algebra | System of Linear Equation | |

$$y = 2x + 7$$

The equation above is line $l$ in the xy plane. Another line $k$, has a slope equal to double the slope of $l$ and a y intercept equal to double the y intercept of line $l$. At which point do the line $l$ and $k$ intersect?

a. $(-\frac{7}{2}, 0)$
b. $(-\frac{2}{7}, 0)$
c. $(0, \frac{2}{7})$
d. $(0, \frac{7}{2})$


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| | | | |
|---|---|---|---|
| 8 | Advanced | Quadratic Function | |
| | The graphs in the xy-plane of the following quadratic equation each have x-intercepts of $-2$ and 4. The graph of which equation has its vertex farthest from the x-axis?<br>a. $y = -10(x+2)(x-4)$<br>b. $y = 8(x+2)(x-4)$<br>c. $y = -\frac{1}{3}(x+2)(x-4)$<br>d. $y = \frac{1}{10}(x+2)(x-4)$ | | |
| 9 | Advanced Math | Exponential Function Word Problem | |
| | The population of a colony if bacteria increases by 300% every 8 hours. If the current population is 5,000, which of the expression represent the colony's population h hours from now?<br>a. $5,000(3)^{\frac{h}{8}}$<br>b. $5,000(4)^{h}$<br>c. $5,000(4)^{8h}$<br>d. $5,000(4)^{\frac{h}{8}}$ | | |
| 10 | Advanced Math | Quadratic | |
| | $y = x^2 + 3$<br>$y = 15x - 33$<br>The system of equation above is graphed in the xy plane. If the system has two solution, what is the product of the x coordinate of the two solutions?<br>a. 36<br>b. 4<br>c. -4<br>d. -36 | | |
| 11 | Advanced Math | Rational Expression | |
| | What is the smallest positive integer value of x such that $\frac{6}{x} + \frac{1}{2x}$ is less than 2? | | |
| 12 | Advanced Math | Exponential function | |

| $x$ | 0 | 1 | 2 |
|---|---|---|---|
| $f(x)$ | 9 | 12 | 16 |

For the exponential function $f$, the table show some values of $x$ dan corresponding of $f(x)$. The function can be written in the form of $f(x) = pr^2$, where $p$ and $r$ are constants. What is the value of $r$?

| | | | |
|---|---|---|---|
| 13 | Advanced Math | Word Problem | |
| | $L = S\sqrt{1 - \frac{v^2}{c^2}}$ | | |



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When a speed of an object approached the speed of light, its length as seen by an observer changes. When the object is stationary relative to an observer, its length is S and when the same object is moving at speed v relative to observer, its lengths is L. The formula above expresses L in term of S, v and c, the speed of light. Which the following gives the speed of the object in the term other quantities?

a. $v = c\sqrt{1 - \frac{L^2}{S^2}}$

b. $v = c\sqrt{1 + \frac{L^2}{S^2}}$

c. $v = c^2\left(1 - \frac{L^2}{S^2}\right)$

d. $v = c^2\left(1 + \frac{L^2}{S^2}\right)$

| 14 | Advanced Math | Polynomial | |
|---|---|---|---|

One of factors of $2x^3 + 14x^2 + 24x$ is $x + a$ , where $a$ is a positive constant. What is the product of the possible number $a$ ?

| 15 | Problem Solving and Data Analysis | Rate and Unit | |
|---|---|---|---|

An automobile uses 27 pints of fuel for ever 63 miles traveled. How many pint of fuel does the automobile use to travel 7 miles?

a. 3
b. 4
c. 9
d. 16

| 16 | Problem Solving and Data Analysis | Probability | |
|---|---|---|---|

The table summarizes the number of public schools in two California counties in 2017.

| | County | | |
|---|---|---|---|
| **School** | Los Angeles | San Diego | Total |
| Elementary | 1,395 | 498 | 1,893 |
| Middle | 422 | 165 | 587 |
| High | 570 | 191 | 761 |
| Total | 2,387 | 854 | 3,241 |

A public middle school will be selected at random from the two counties. What is the probability, to the nearest hundredth, of selecting a school in San Diego County?

a. 0.05
b. 0.19
c. 0.28
d. 0.69

| 17 | Problem Solving and Data Analysis | Statistic | |
|---|---|---|---|

A set of data has mean of 15 and range 6. If every value of the data is subtracted by A and then the result is divided by B, it produce a new data set with mean of 7 and range of 3. What are the value of A and B, respectively?



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a. 3 and 2
b. 2 and 3
c. 1 and 2
d. 2 and 1

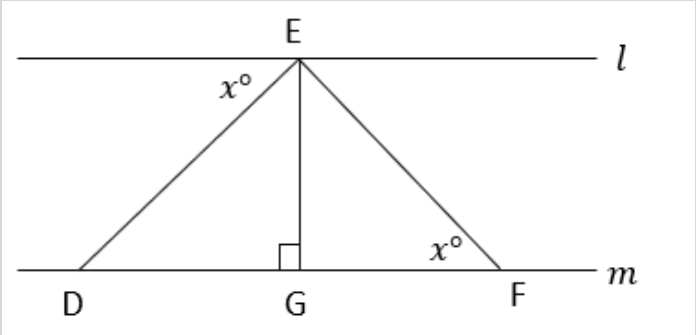
| 18 | Problem Solving and Data Analysis | Statistic | |
|---|---|---|---|

| Value | Frequency |
|---|---|
| 1 | 1 |
| 2 | 4 |
| 3 | 2 |
| 4 | 3 |
| 5 | 7 |

What is the median of the data set summarized in the frequency table?

a. 2
b. 3
c. 4
d. 5

| 19 | Geometry | Line and Angle | |
|---|---|---|---|



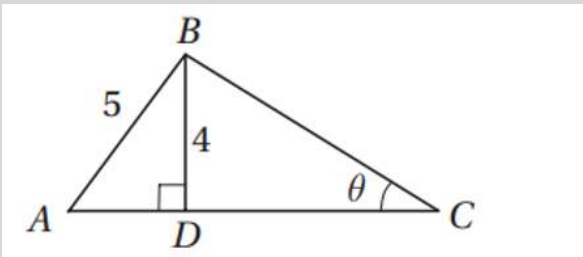

In the figure above, line $l$ is parallel to line $m$. If $x = 40$ what is the measure of $\angle DEF$?

a. 140°
b. 100°
c. 80°
d. 50°

| 20 | Geometry | Triangle | |
|---|---|---|---|

What is the perimeter of an equilateral triangle with a height $10\sqrt{3}$ ?

| 21 | Geometry | Trigonometry | |
|---|---|---|---|





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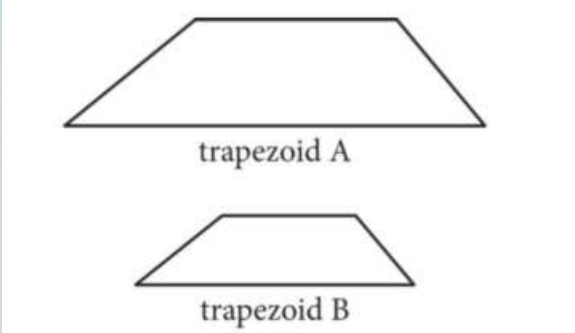
| | | | |
|---|---|---|---|
| | In the figure above, triangle ABC has area of 19. What is the value of tan $tan$ θ ? [Answer in a/b where a and b are integer] | | |
| 22 | Geometry | Area of similar figure | |




Trapezoid A and trapezoid B shown are similar. The length of each side of trapezoid A is 16 time of the length of corresponding side of trapezoid B. The area trapezoid B is how many times as large as the area of trapezoid A?

a. $\frac{1}{256}$
b. $\frac{1}{4}$
c. 4
d. 256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